

Sensor Current Requirement for Floating Phytoremediation & Hydroponic Robot

This document summarizes the estimated current requirements for all sensors used in the floating robot integrating phytoremediation, hydroponics, and water cleaning functionalities. All values are based on typical operating parameters of commercially available modules such as Atlas Scientific probes, u-blox GPS, and Raspberry Pi camera modules.

1. Sensor Current Requirements

Sensor	Operating Voltage (V)	Current Draw (mA)	Notes / Duty
pH Probe + Interface	5	20	Continuous
EC / TDS Probe	5	20	Continuous
Dissolved Oxygen (Optical Probe)	5	60	Continuous
ORP Sensor	5	15	Continuous
Turbidity Sensor	5	10 (≈ 1 avg)	10% duty
Temperature Sensor	3.3	1.5	Continuous
GPS Module	3.3	60	Continuous
IMU (Accelerometer/Gyro)	3.3	3.5	Continuous
Ultrasonic Rangefinder	5	10 (≈ 1 avg)	10% duty
Depth / Pressure Sensor	3.3	1	Continuous
Camera Module	5	300 (≈ 50 avg)	16% duty
Spectrometer / Optical Sensor	3.3	60 (≈ 6 avg)	10% duty

2. Total Estimated Current Draw

- Continuous sensors total ≈ 181 mA
- Intermittent sensors (average) ≈ 58 mA
- Grand total (average across all sensors): ≈ 239 mA

Note: Because the system uses multiple voltage levels (5 V and 3.3 V), DC-DC conversion losses (approx. 90% efficiency) must be accounted for. The resulting current drawn from the main 12 V supply will be slightly higher.

3. Practical Notes

- Design for 20–30% current margin to accommodate calibration and startup surges.
- Regular sensor calibration is essential for pH, EC, and DO sensors.
- Employ anti-fouling mechanisms (e.g., mechanical wipers, flush cycles) for optical sensors.
- If using solar power, size the battery for 2–3 days of autonomy based on total daily energy usage.

Floating Robot — Phytoremediation + Hydroponics Block Diagram

